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## 1 PSOC™ DevCon Americas Hands-on Courses requirements

### Instructions:

1. Check tables of [ModusToolbox™ requirements](#) and [additional tools](#), and install ALL the tools for the courses and tracks you are attending to before the event
2. Complete [pre-requisite courses](#) before the event

### For any questions or issues regarding, contact:

- PSOC™ Edge:
  - Luis Reynoso, [Luis.Reynoso@infineon.com](mailto:Luis.Reynoso@infineon.com)
- PSOC™ Control:
  - Jahangir Mansoori, [Jahangir.Mansoori@infineon.com](mailto:Jahangir.Mansoori@infineon.com)
- PSOC™ Multi-Sense:
  - Kyle Reynolds, [Kyle.Reynolds@infineon.com](mailto:Kyle.Reynolds@infineon.com)
- Zephyr and MTB Advanced:
  - Clark Jarvis, [Clark.Jarvis@infineon.com](mailto:Clark.Jarvis@infineon.com)

## 2 Install or Update ModusToolbox™ and MTB tools

Use this table to check ModusToolbox requirements for the DevCon tracks which you are participating in.

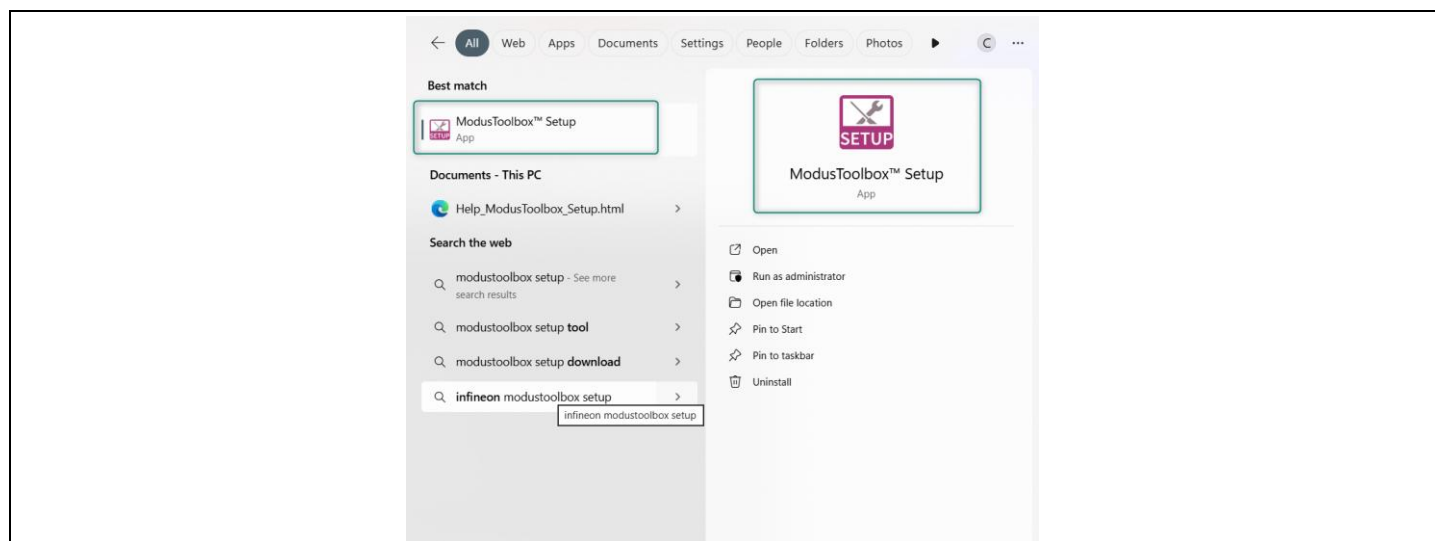
| Tool                                         |           | Version (or later) | PSOC™ Control track | PSOC™ Edge track | PSOC™ Multi-Sense track | Gold training | Zephyr | General session-<br>ModusToolbox advanced |
|----------------------------------------------|-----------|--------------------|---------------------|------------------|-------------------------|---------------|--------|-------------------------------------------|
| ModusToolbox Tools Package                   | 3.9       | ✓                  | ✓                   | ✓                | ✓                       |               | ✓      |                                           |
| Eclipse IDE for ModusToolbox                 | 2026.3.0  | ✓                  |                     |                  | ✓                       |               |        |                                           |
| ModusToolbox Programming Tools               | 1.9.0     | ✓                  | ✓                   | ✓                | ✓                       |               | ✓      |                                           |
| Arm GNU Toolchain (GCC)                      | 14.2.1    | ✓                  | ✓                   | ✓                | ✓                       |               | ✓      |                                           |
| ModusToolbox Edge Protect Security Suite     | 2.2.0     |                    | ✓                   |                  | ✓                       |               | ✓      |                                           |
| DEEPCRAFT Audio Enhancement Tech Pack        | 1.3.0     |                    | ✓                   |                  | ✓                       |               |        |                                           |
| ModusToolbox Audio SW Codecs Tech Pack       | 1.0.3     |                    | ✓                   |                  | ✓                       |               |        |                                           |
| ModusToolbox CAPSENSE and Multi-Sense        | 1.6       |                    |                     | ✓                | ✓                       |               |        |                                           |
| ModusToolbox Motor Suite                     | 2.9       | ✓                  |                     |                  | ✓                       |               |        |                                           |
| ModusToolbox Power Suite                     | 1.2.0*    | ✓                  |                     |                  | ✓                       |               |        |                                           |
| ModusToolbox LiveMonitoring                  | 1.0.x     | ✓                  |                     |                  | ✓                       |               |        |                                           |
| ModusToolbox Machine Learning Pack           | 3.3.0     |                    | ✓                   |                  | ✓                       |               |        |                                           |
| DEEPCRAFT Studio                             | 5.14.5788 |                    | ✓                   |                  | ✓                       |               |        |                                           |
| DEEPCRAFT ModelConverter                     | 5.11.5765 |                    | ✓                   |                  | ✓                       |               |        |                                           |
| *Expected to be released beginning of Oct-26 |           |                    |                     |                  |                         |               |        |                                           |

### 2.1 Install or Update ModusToolbox™ v3.9.0

If the ModusToolbox™ Software is not installed, please download the software from the [Infineon website](#) and follow the [Installation Guide](#).

If ModusToolbox™ is already installed, confirm the version is v3.9 or higher:

- Open ModusToolbox™ Setup:



**Figure 1 Open ModusToolbox™ Setup**

- Verify the installed version is v3.9.0 or higher:

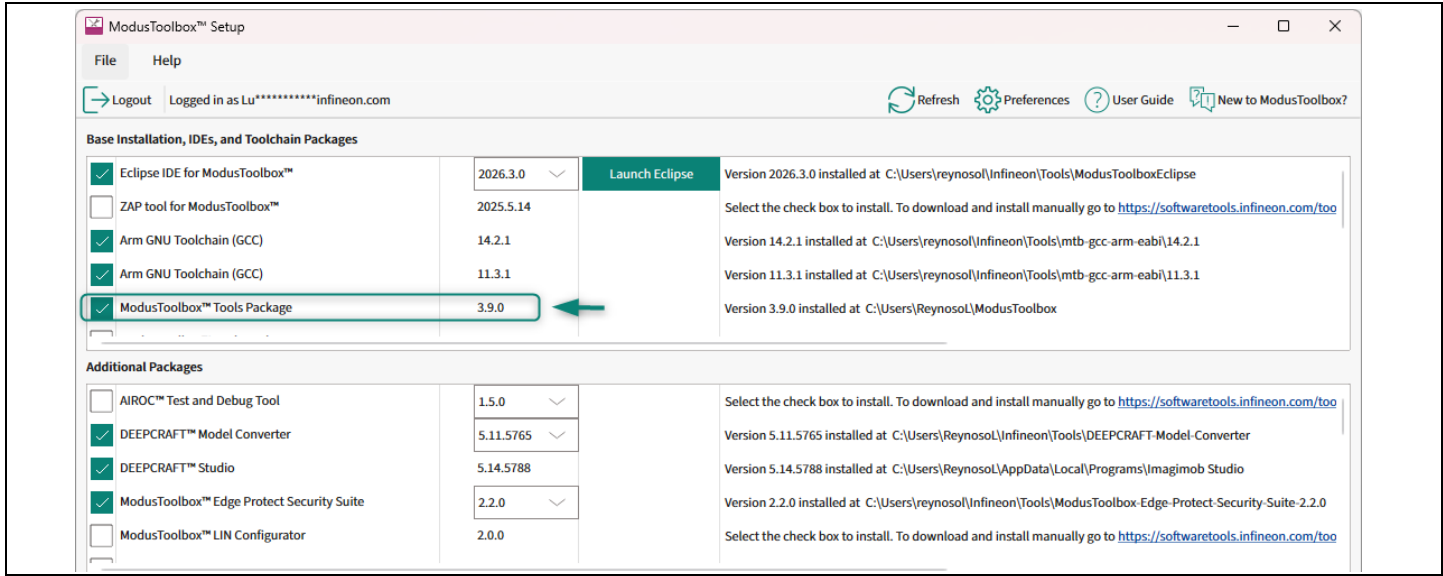
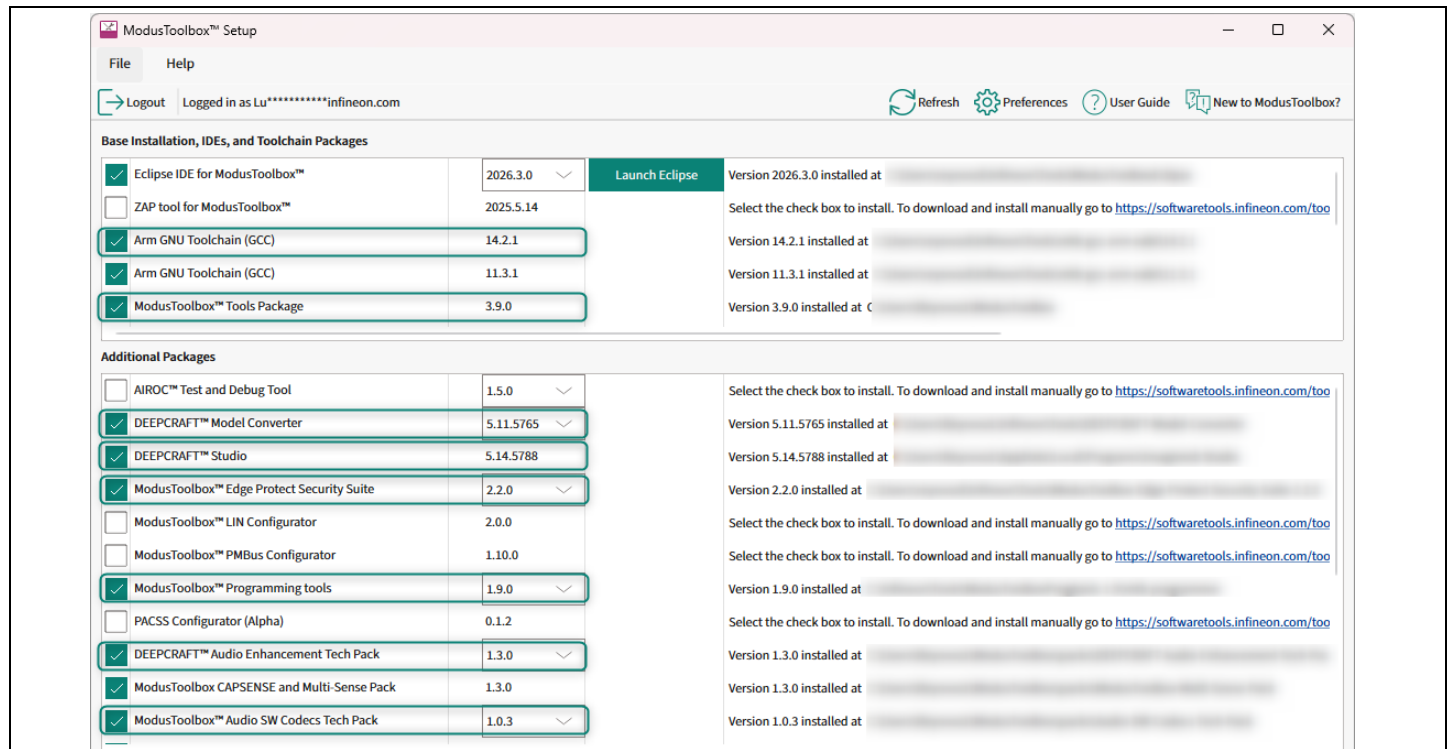


Figure 2 ModusToolbox™ version

## 2.2 Install or Update ModusToolbox™ Packages as required for your courses

Using the table at the front of this document, install or update **all** of the ModusToolbox™ packages for your courses.



**Figure 3 Example: ModusToolbox™ Packages for PSOC™ Edge Training Labs**

### Tools can also be installed from direct links:

- ModusToolbox: <https://softwaretools.infineon.com/tools/com.ifx.tb.tool.modustoolbox>
- Eclipse IDE: <https://softwaretools.infineon.com/assets/com.ifx.tb.tool.mtbeclipse>
- Programming tools: <https://softwaretools.infineon.com/assets/com.ifx.tb.tool.modustoolboxprogttools>
- Arm GNU Toolchain: <https://softwaretools.infineon.com/assets/com.ifx.tb.tool.mtbgccpackage>
- Edge Protect Security Suite: <https://softwaretools.infineon.com/assets/com.ifx.tb.tool.modustoolboxedgeprotectsecuritysuite>
- DEEPCRAFT audio enhancement tech pack: <https://softwaretools.infineon.com/assets/com.ifx.tb.tool.deepcraftaudioenhancementtechpack>
- Audio SW codecs tech pack: <https://softwaretools.infineon.com/tools/com.ifx.tb.tool.modustoolboxpackaudioswcodecs>
- CAPSENSE and Multi-Sense pack: <https://softwaretools.infineon.com/assets/com.ifx.tb.tool.modustoolboxpackmultisense>
- Motor Suite: <https://softwaretools.infineon.com/tools/com.ifx.tb.tool.ifxmotorsolutions>
- Power Suite: <https://softwaretools.infineon.com/tools/com.ifx.tb.tool.modustoolboxpowersuite>
- Live Monitor: <https://softwaretools.infineon.com/tools/com.ifx.tb.tool.modustoolboxpacklivemonitor>
- Machine learning pack: <https://softwaretools.infineon.com/tools/com.ifx.tb.tool.modustoolboxpackmachinelearning>
- DEEPCRAFT Studio: <https://softwaretools.infineon.com/assets/com.ifx.tb.tool.deepcraftstudio>
- DEEPCRAFT Model Converter: <https://softwaretools.infineon.com/assets/com.ifx.tb.tool.deepcraftmodelconverter>

### 3 Additional tools

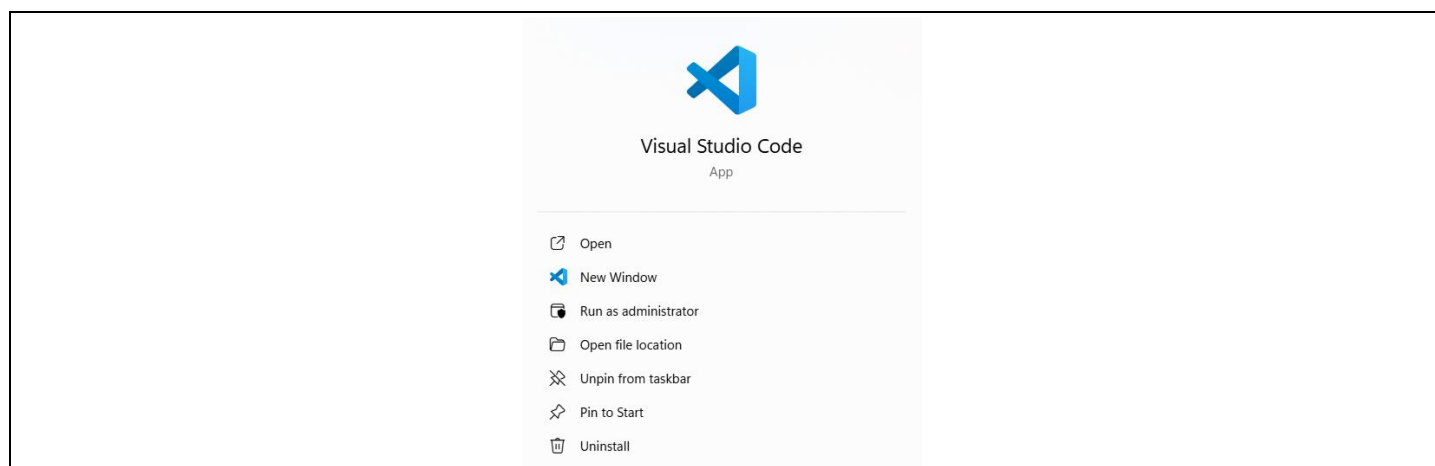
| Tool                                 | Version (or later) | PSOC™ Control track | PSOC™ Edge track | PSOC™ Multi-Sense track | Gold training | Zephyr   | General session-<br>ModusToolbox advanced |
|--------------------------------------|--------------------|---------------------|------------------|-------------------------|---------------|----------|-------------------------------------------|
| Microsoft Visual Studio Code         | 2026.1.0           | ✓                   | ✓                | ✓                       |               | ✓        |                                           |
| Tera Term, or other serial terminal  | -                  | ✓                   | ✓                | ✓                       |               | ✓        |                                           |
| Git for Windows                      | 2.55               |                     |                  |                         |               | ✓        |                                           |
| Zephyr dependencies                  | -                  |                     |                  |                         | ✓             |          |                                           |
| PSoC Programmer                      | 3.29.6.4838        |                     | ✓                |                         |               |          |                                           |
| J-Link Driver                        | v9.74              | ✓                   |                  | ✓                       |               |          |                                           |
| LLVM ARM Compiler                    | 19.1.5             | ✓                   |                  | ✓                       |               |          |                                           |
| Audacity                             | v3.7.5             | ✓                   |                  |                         |               |          |                                           |
| EEZ-Studio                           | v0.29.0            | ✓                   |                  | ✓                       |               |          |                                           |
| Python3 with pypng and lz4 modules   | v3.12              | ✓                   |                  | ✓                       |               |          |                                           |
| Pngquant                             | v2.17.0            | ✓                   |                  | ✓                       |               |          |                                           |
| DEEPCRAFT Voice Assistant account    | -                  | ✓                   |                  | ✓                       |               |          |                                           |
| Nmap                                 | v7.98              | ✓                   |                  |                         |               |          |                                           |
| Wireshark                            | v4.6.3             | ✓                   |                  |                         |               |          |                                           |
| GitHub Copilot or other AI assistant | -                  | Optional            | Optional         | Optional                | Optional      | Optional | Optional                                  |

#### 3.1 Tools shared across multiple tracks

##### 3.1.1 Install Microsoft Visual Studio Code

[Microsoft Visual Studio Code](#) is a modern IDE that allows for extensions such as the [ModusToolbox™ for VS Code](#) to be used to build, program, and debug embedded firmware projects for PSOC™ devices.

Once installed, launch Microsoft VS Code from the Windows search bar.



**Figure 4** Visual Studio Code in Windows search bar

Once you open VS Code, click on the **Extensions** icon, then search for “Infineon”. Click on **Infineon Modustoolbox™ for VS Code**, then in the main view click the install button.

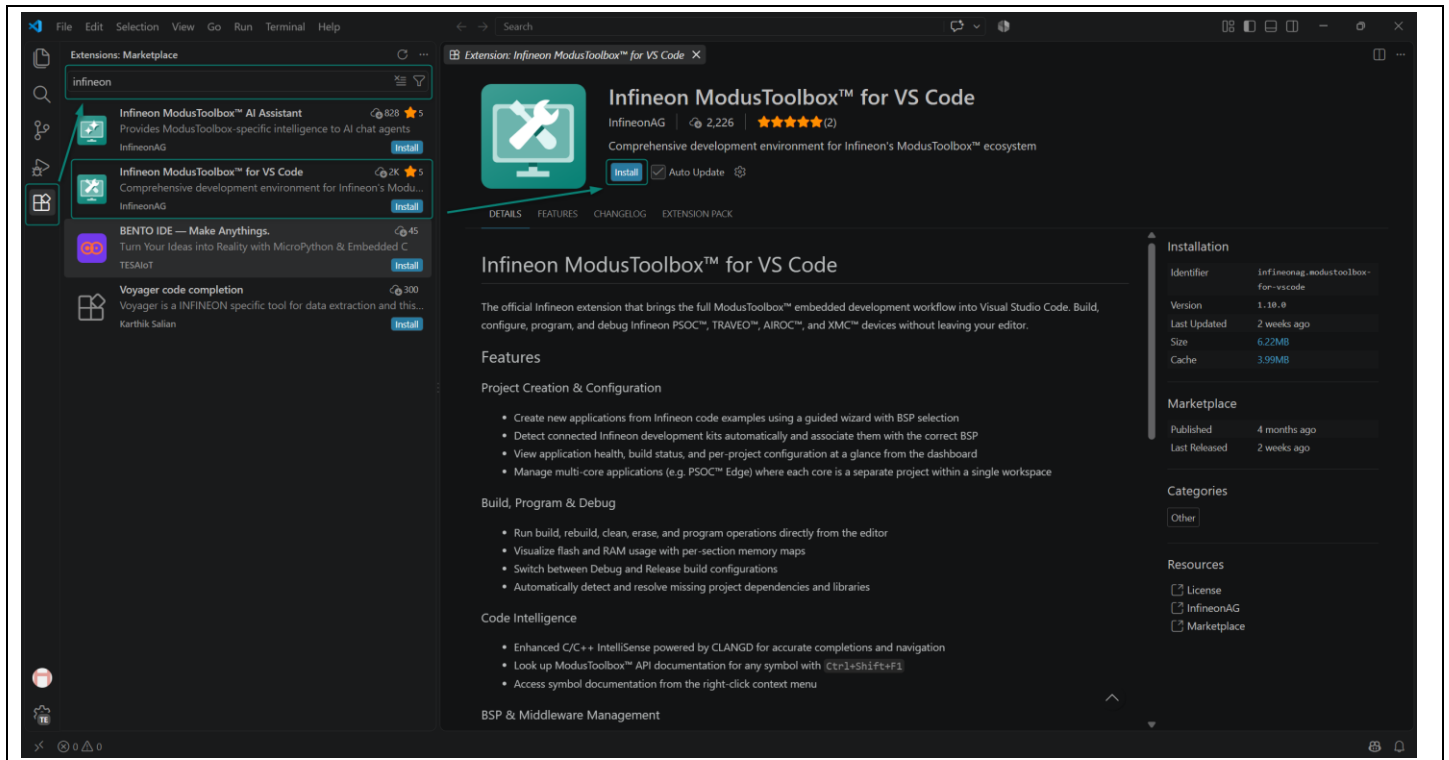


Figure 5 Installing Modustoolbox™ for VS Code extension

Once installed, press **ctrl+shift+p** to open the command palette. Then type **Infineon Modustoolbox™ for VS Code: Show Main Page**. Click on the command to execute it and the extensions main page will open. An icon in the top right will be available to take you back to the extension as long as the extension is running.

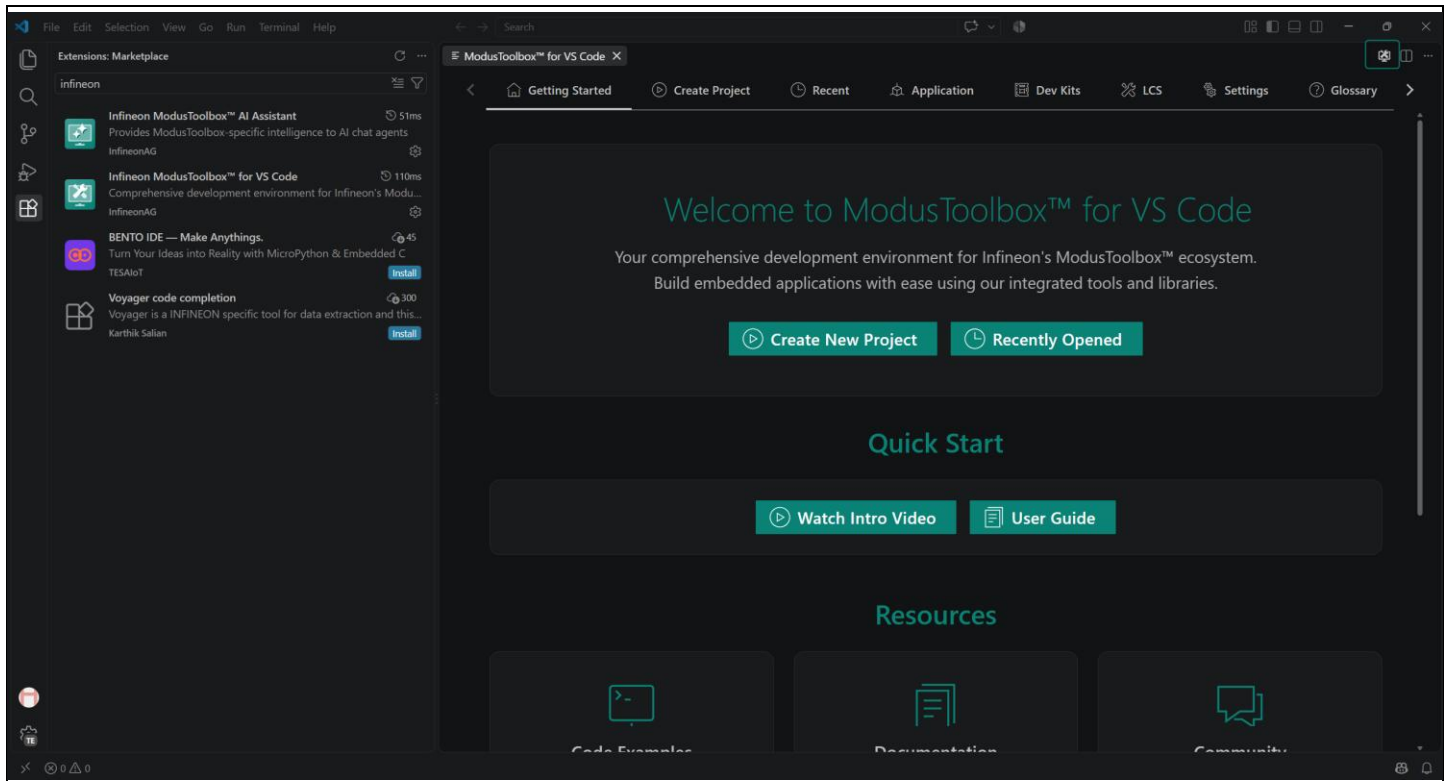


Figure 6 Installing Modustoolbox™ for VS Code extension

### 3.1.2 Install a serial terminal emulator (e.g. Tera Term)

A serial terminal will be used for some labs. [Tera Term](#) is just one recommendation. Visual Studio Code also has an extension for a [serial monitor](#).

### 3.1.3 Install Git for Windows

Note: Git for Windows will be installed as a [Zephyr dependency](#). It is not necessary to install it twice.

1. Download and install Git for Windows from <https://git-scm.com/install/windows>
2. When installing, make sure Git is added to the PATH

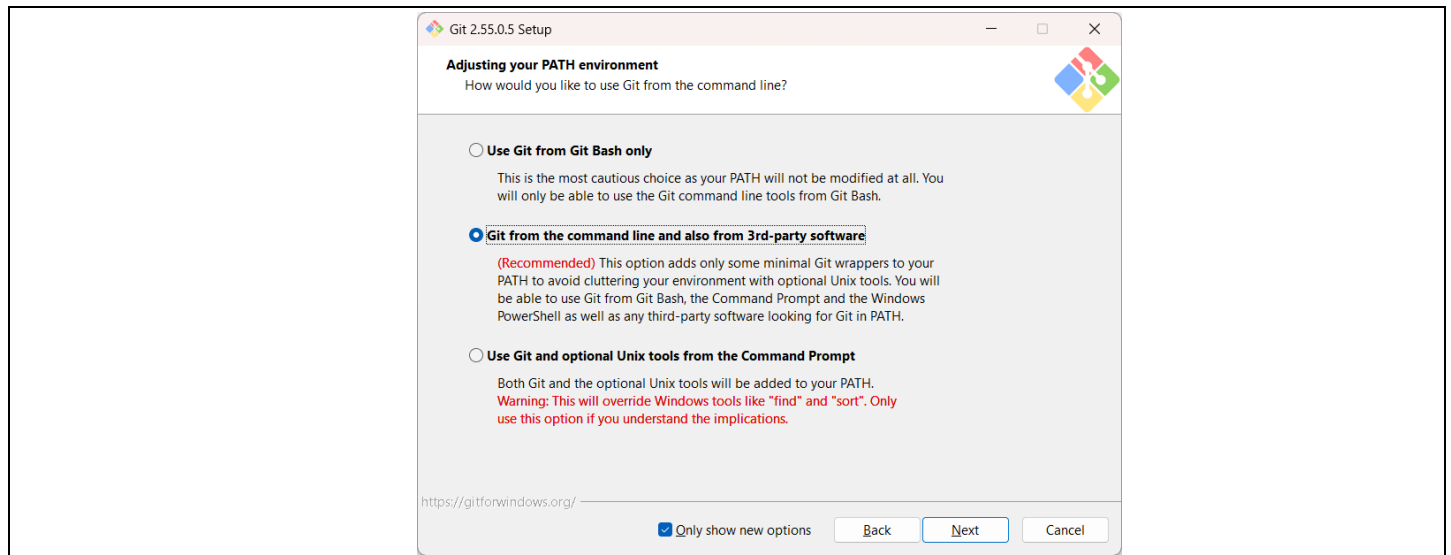


Figure 7 Add Git to PATH

3. Confirm installation by opening a command prompt and typing:

➤ `git --version`

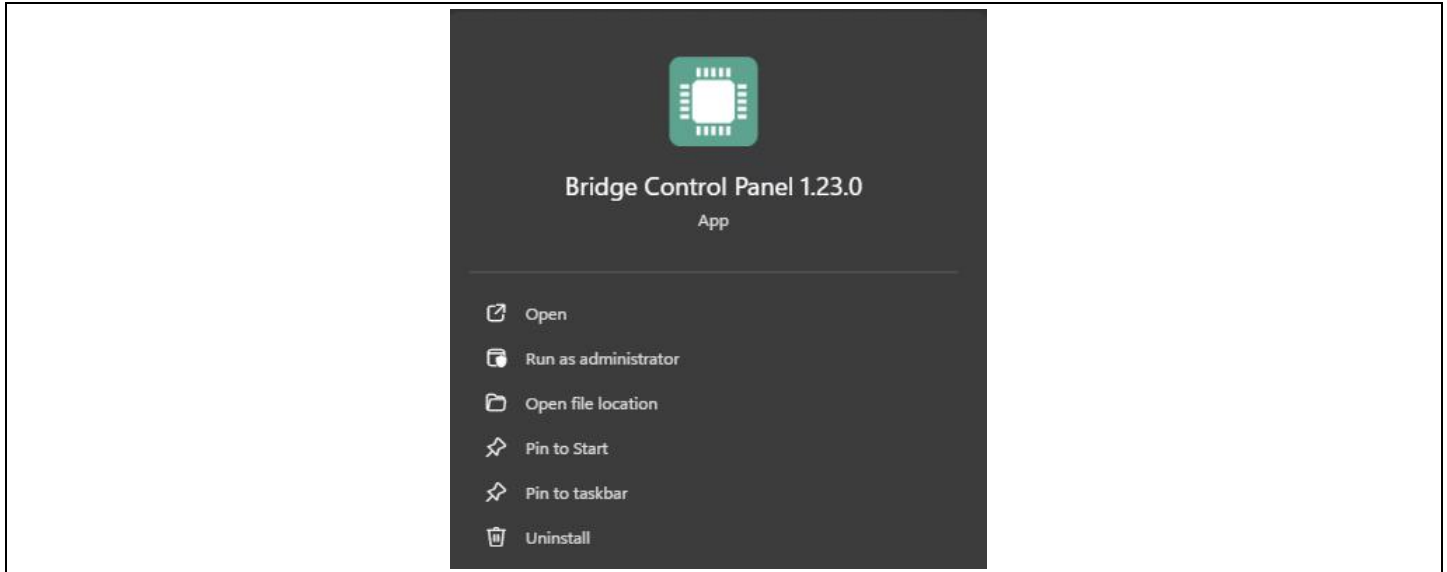
## 3.2 Additional tools for PSoC™ Multi-Sense

### 3.2.1 Install PsoC™ Programmer

[PSoC™ Programmer](#) installs tools for programming and debugging PSoC™ devices. The PSoC™ Multi-Sense track will use the **Bridge Control Panel**, which is installed with the PSoC™ Programmer.

Once installed, the **Bridge Control Panel** should be accessible from the Windows search bar





**Figure 8 Bridge Control Panel in Windows search bar**

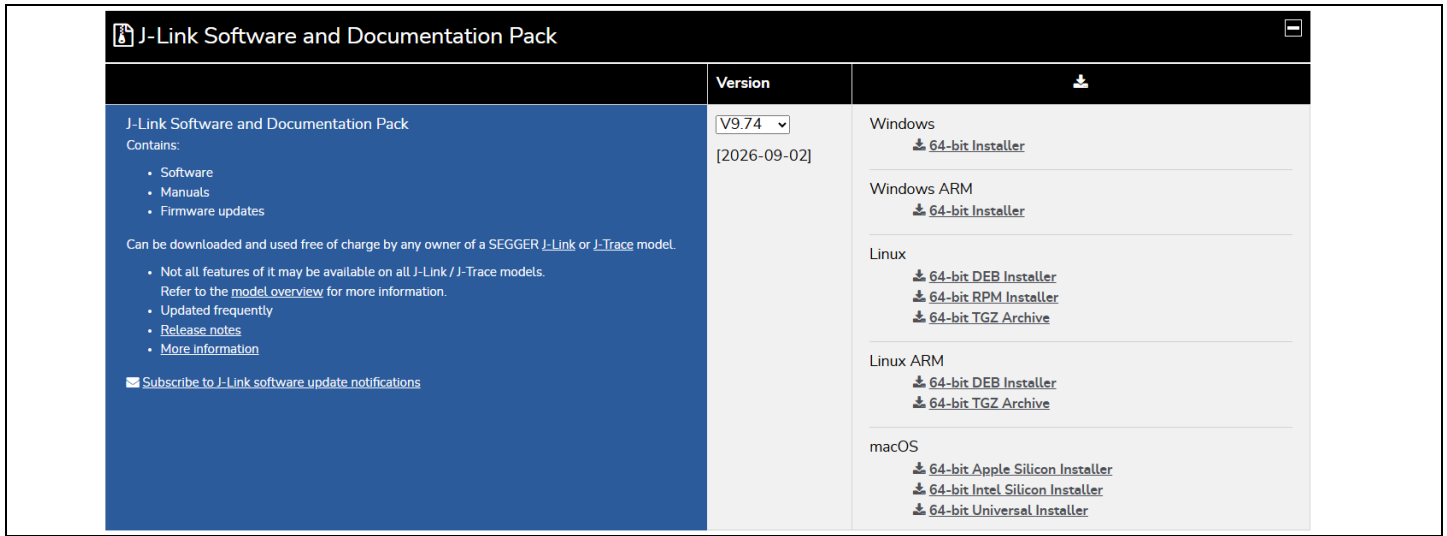
### 3.3 Additional tools for PSOC™ Control

#### 3.3.1 J-Link Driver installation

J-Link drivers are needed to program and debug some kits, today the following kits are supported by J-Link:

- KIT\_PSC3Mx\_CCy, with x = [5, 6, 8] and with y = [1 or 2]
- KIT\_PSC3M6\_EVK

1. Go to <https://www.segger.com/downloads/jlink/>
2. Select the latest J-Link driver version
3. Pick the installer according to your operating system



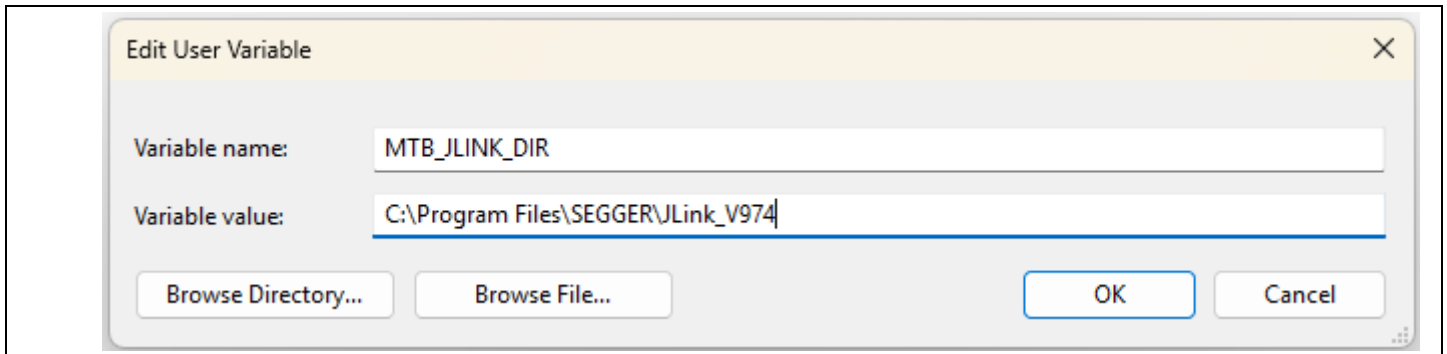
**Figure 9 Download J-Link drivers**

4. Wait until the installation is complete.

- When accepting the default installation flow provided by SEGGER, the J-link software will be added to the folder retaining the version number: **JLink\_V974**.
- Open the environment variable window and add or edit the user variable to point to this J-Link folder:

**Variable name:** MTB\_JLINK\_DIR

**Variable value:** C:\Program Files\SEGGER\JLink\_V974 (or follow the number of the version installed)



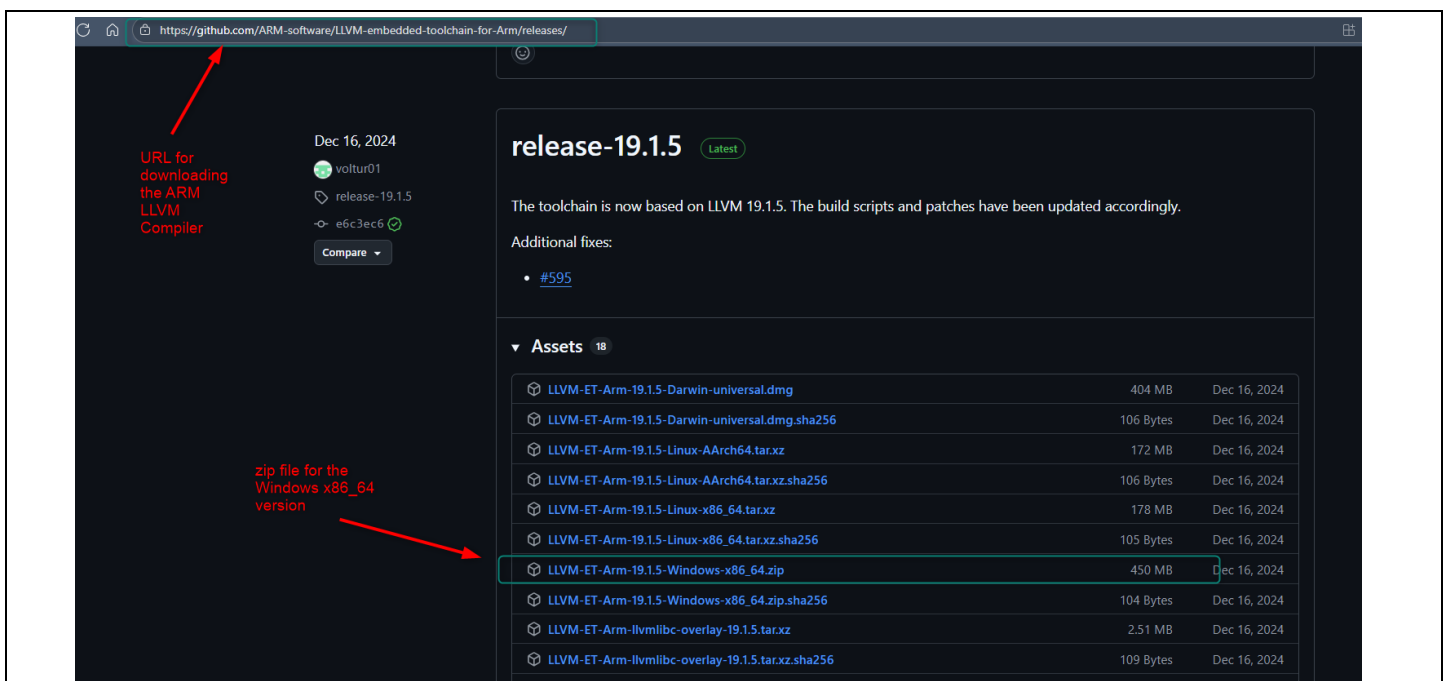
**Figure 10** Add environmental variable for J-Link

## 3.4 Additional tools for PSOC™ Edge

### 3.4.1 Install LLVM ARM Compiler

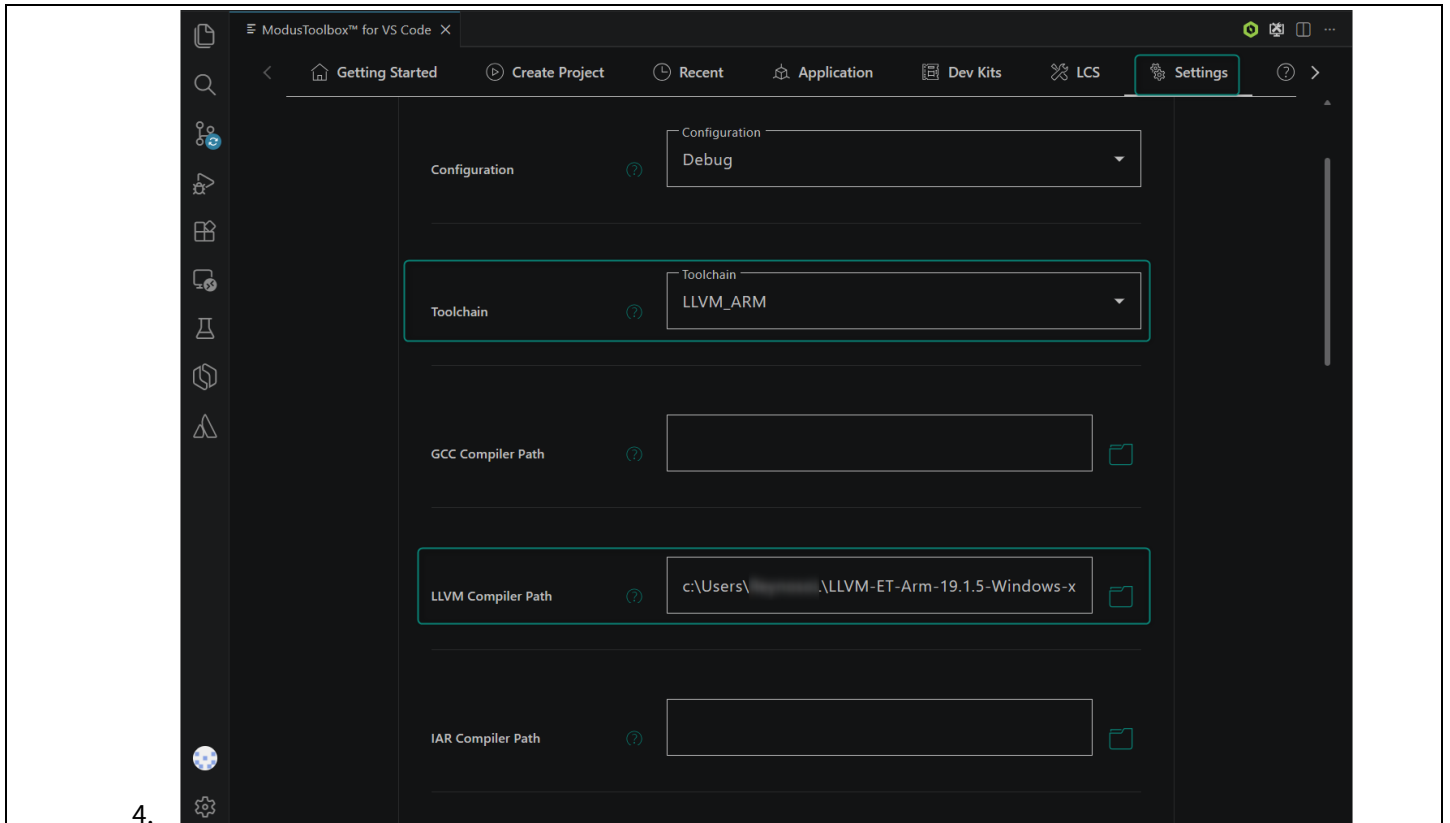
The Arm® LLVM compiler is a free open-source compiler used for some PSOC Edge labs, such as voice and audio.

- Download the Arm® LLVM Compiler Embedded Toolchain v19.1.5 from the Arm® GitHub site:  
<https://github.com/ARM-software/LLVM-embedded-toolchain-for-Arm/releases/>



**Figure 11** Download and Install ARM LLVM Compiler

2. Extract the downloaded zip file into the user folder. i.e.  
`c:\Users\<user>\LLVM-ET-Arm-19.1.5-Windows-x86_64`
3. When using LLVM in Visual Studio Code, open the ModusToolbox extension, select the LLVM toolchain and configure the path accordingly



**Figure 12 Configure LLVM Compiler in VSCode**

### 3.4.2 Install Audacity v3.7.5 or later

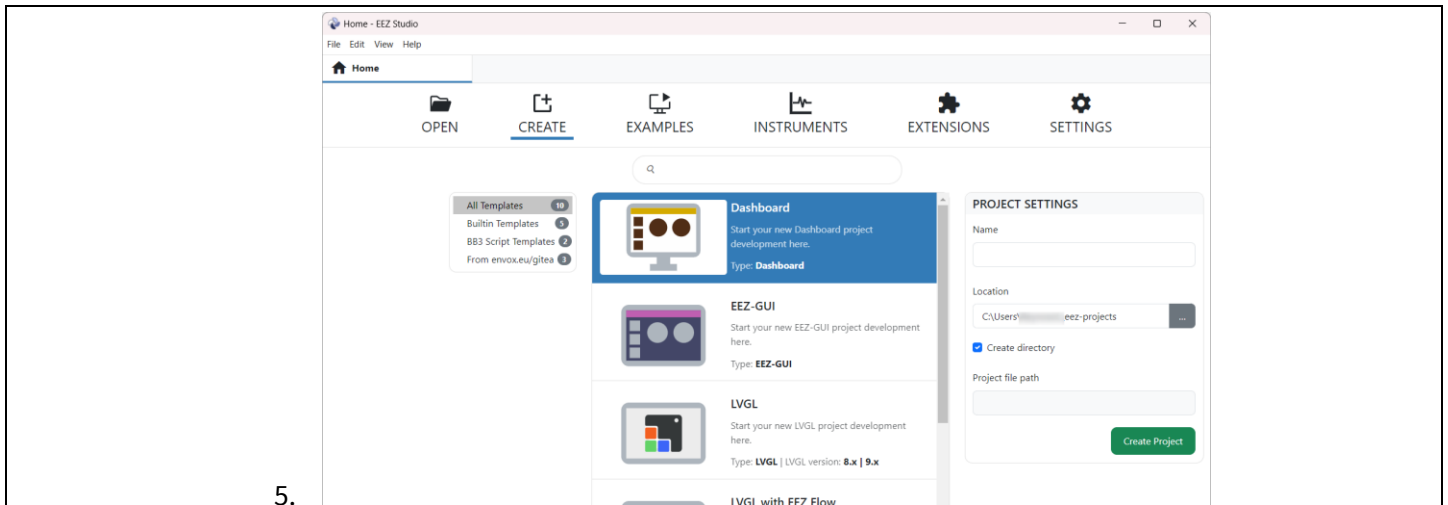
Audacity is a free and open source, cross-platform software for recording and editing sounds. For the audio labs it will be used for audio capture, visualizations, and analysis.

1. Download and install Audacity for your OS:  
<https://www.audacityteam.org/download/>
2. Note: when Audacity starts up for the first time, you may be asked to configure/link a cloud service at audio.com. You do not need this feature.

### 3.4.3 Install EEZ-Studio v0.29.0 or later

EEZ-Studio is required to create a custom UI for PSOC Edge graphics.

1. Visit <https://www.envox.eu/studio/studio-introduction/> and download EEZ-Studio
2. After installing, confirm tool can run

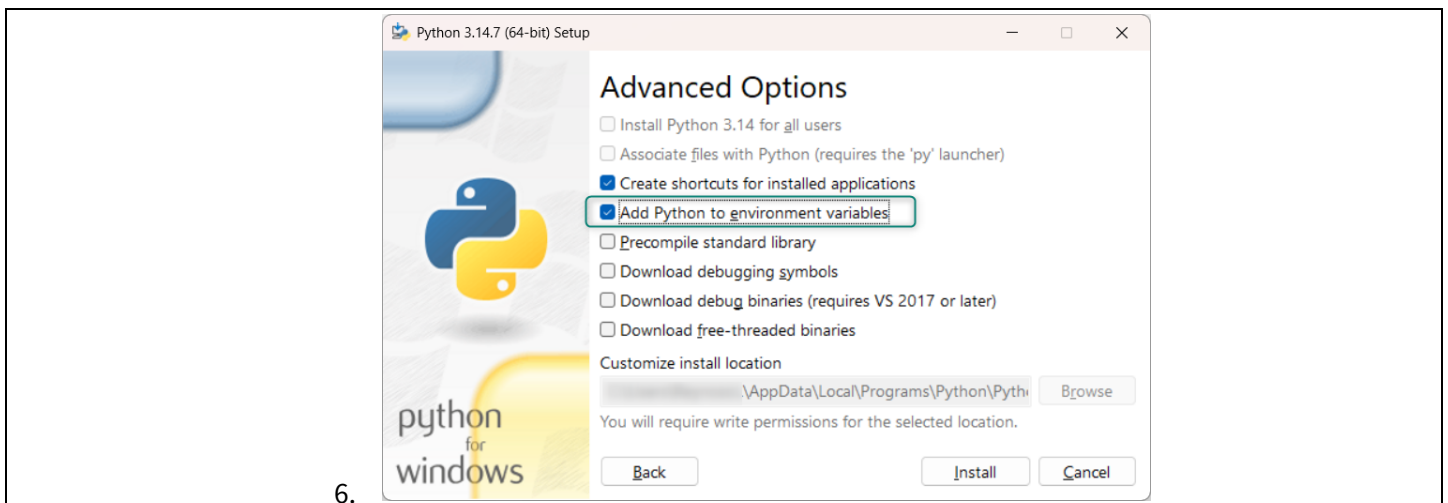


**Figure 13 EEZ-Studio**

### 3.4.4 Install Python3 v0.24.0 or later, with pypng and lz4 modules

Python 3 and modules are only required for graphics labs using EEZ-Studio.

1. Visit <https://www.python.org/downloads/> and install Python3
  - a. When installing, make sure Python is added to environment variables



**Figure 14 Adding Python to environment variables during installation**

2. After installing, open a command prompt to confirm the Python version and installation path by running:
  - `python --version`
  - `where python`

Note: If Python is not found in path, reinstall or [troubleshoot](#).

3. Install “pypng” and “lz4” Python modules. Run the following commands in a command prompt.
  - `python -m pip install pypng==0.20220715.0`
  - `python -m pip install lz4`

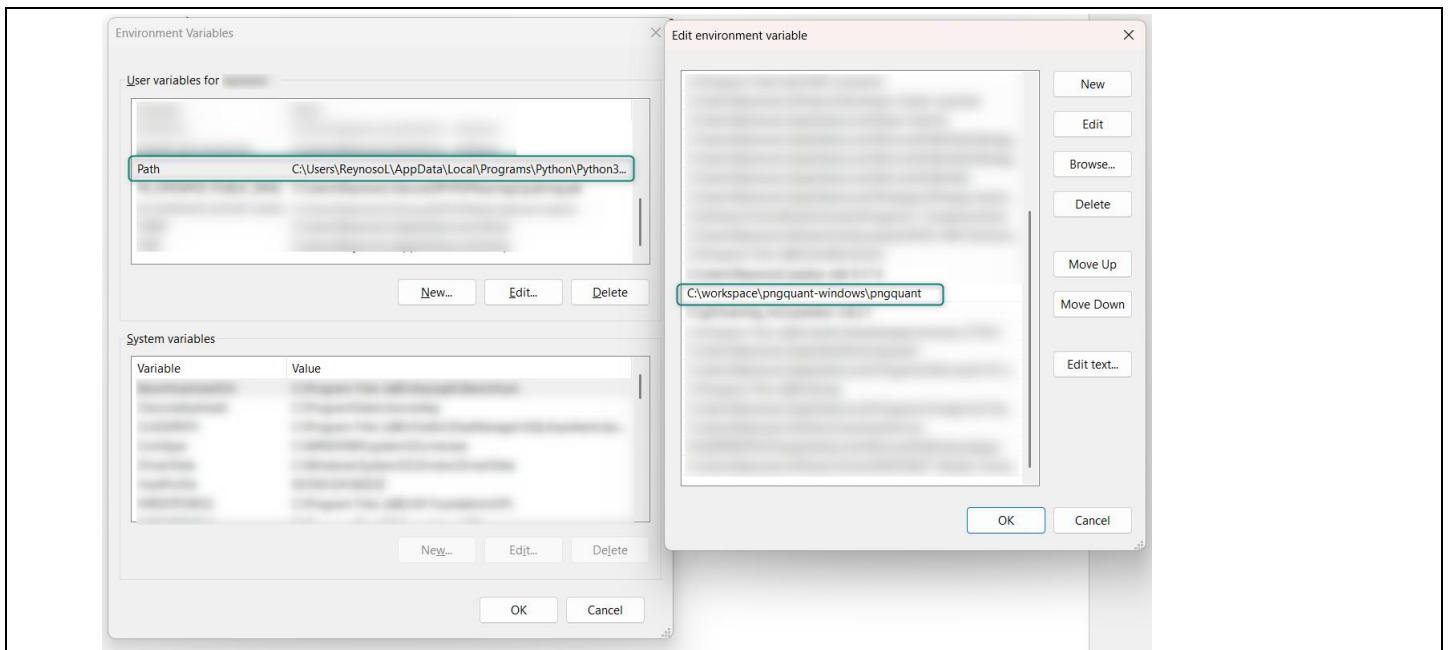
4. Confirm the “pypng” and “lz4” modules are installed.

➤ `pip list`

### 3.4.5 Install pngquant v2.17.0 or later

Pngquant is only required for graphics labs using EEZ-Studio.

1. Visit <https://pngquant.org/>, download and install pngquant
2. After installing, add the pngquant path to the system or user PATH



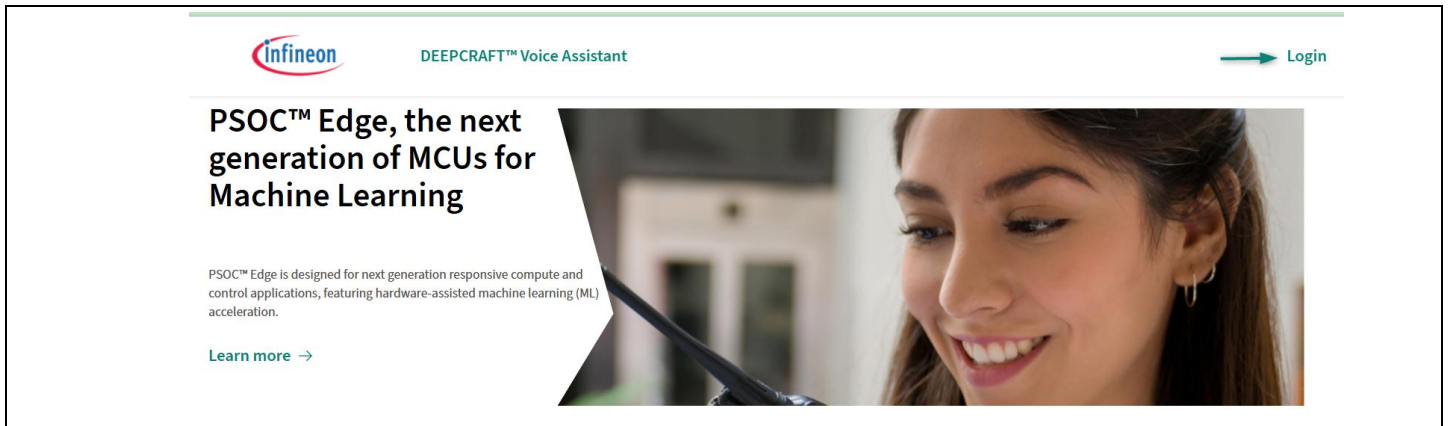
**Figure 15 Adding pngquant to environment variables**

3. To confirm installation, open a command prompt and run:

➤ `pngquant --version`

### 3.4.6 Create DEEPCRAFT™ Voice Assistant account

1. Visit <https://deepcraft-voice-assistant.infineon.com/welcome>
2. Click Login and confirm you can login to the cloud tool, or create an account if necessary.

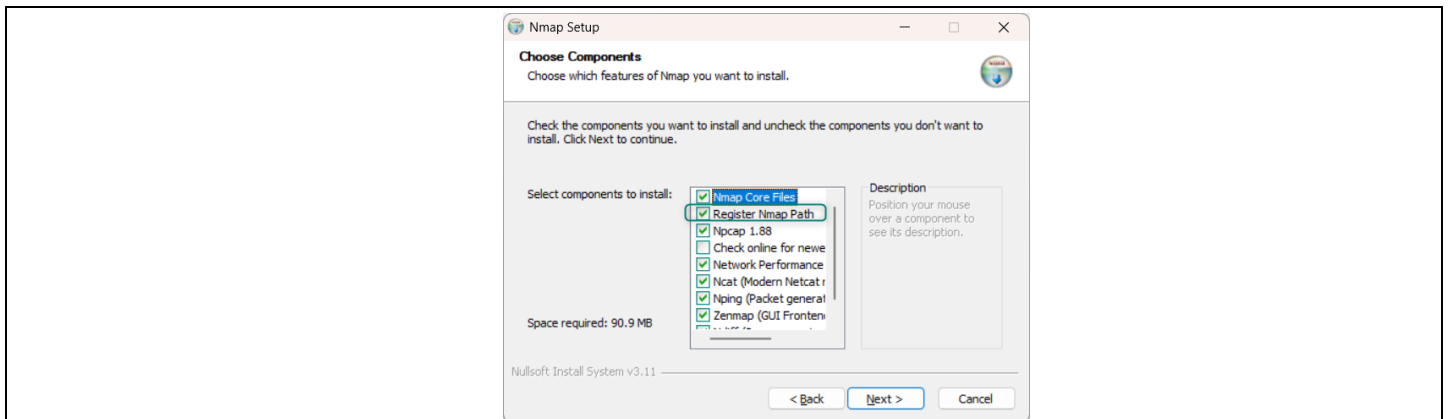


**Figure 16 Create DEEPCRAFT Voice Assistant account**

### 3.4.7 Install Nmap v7.98

Nmap is a free, open-source command-line tool used for network discovery and security auditing. We will be using Nmap for some PSOC Edge Wireless labs.

1. Download and install Nmap v7.98 (or newer) from <https://nmap.org/download> (admin rights needed).
2. When installing select “Register Nmap path” to add Nmap to the system path (default setting).



**Figure 17 Register Nmap path**

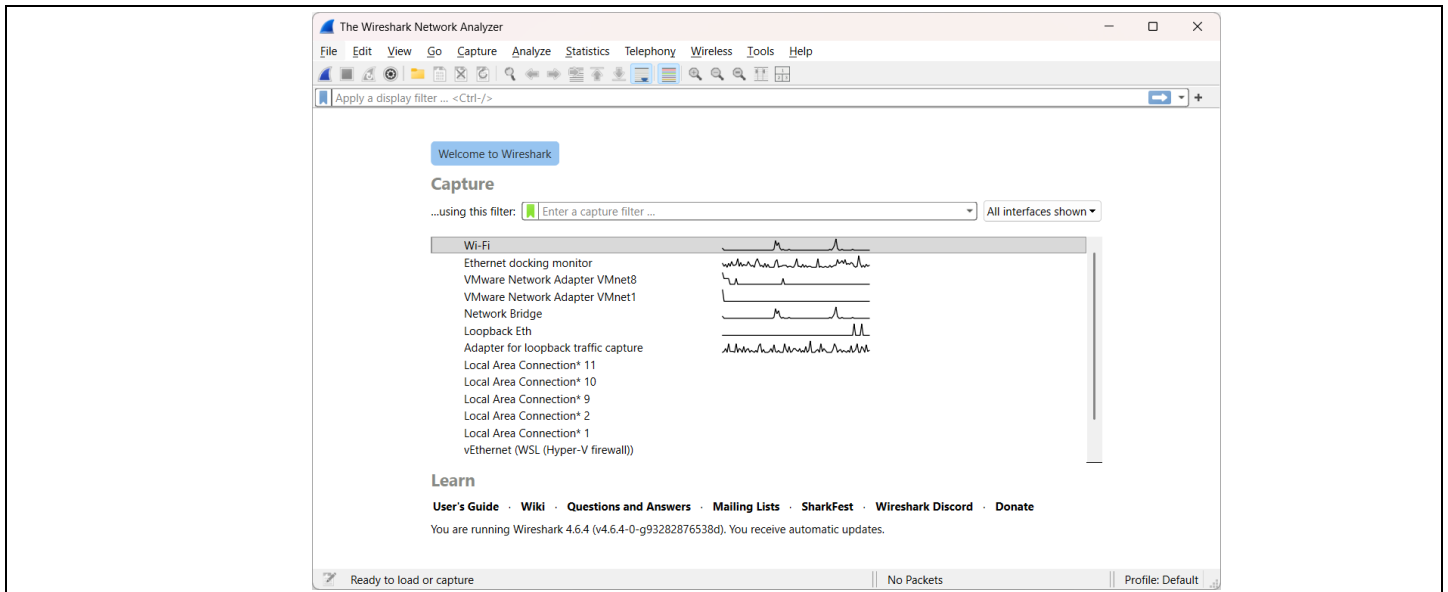
3. Confirm installation by opening a command prompt and typing:

➤ `nmap --version`

### 3.4.8 Install Wireshark v4.6.3

Wireshark is a free computer program that captures and inspects data moving across a network. We will be using Wireshark for some PSOC Edge Wireless labs.

1. Download and install Wireshark v4.6.3 (or newer) from <https://www.wireshark.org/download.html> (admin rights needed).
2. Default installation settings are OK.
3. Confirm installation by opening Wireshark from your OS



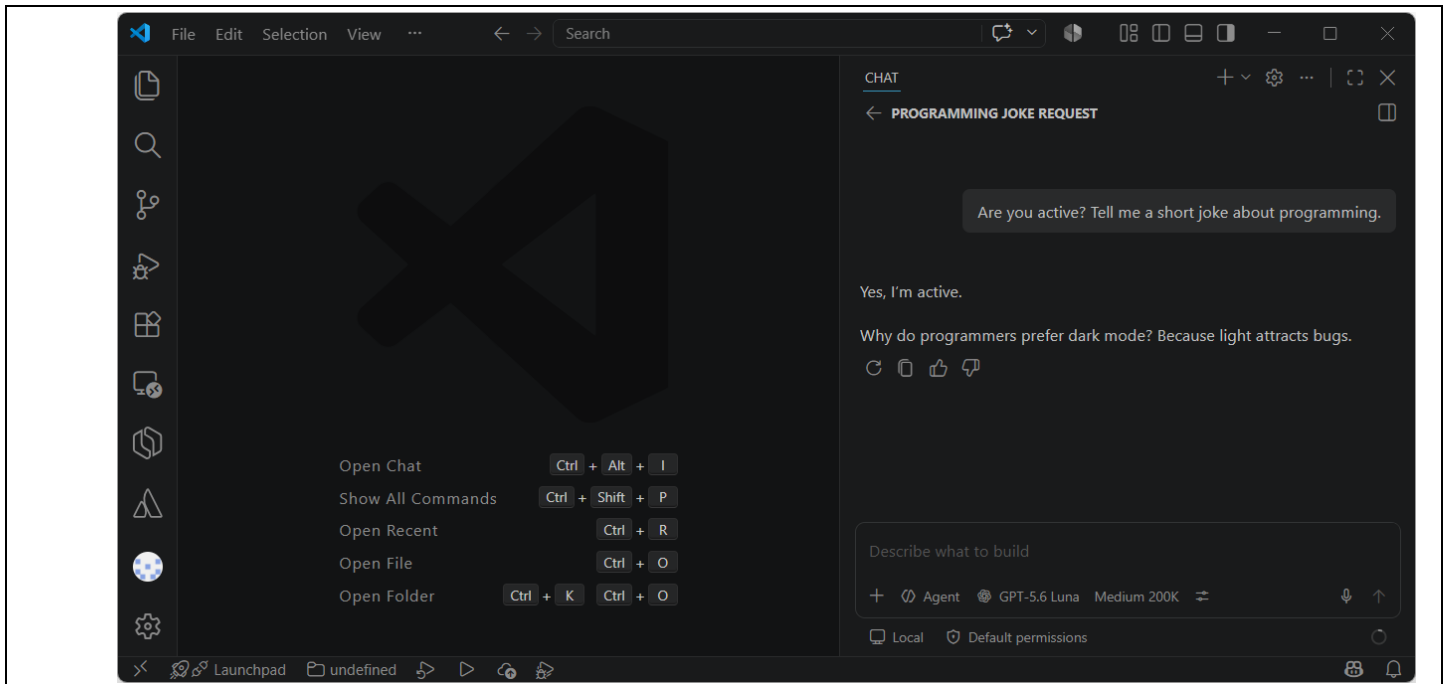
**Figure 18 Running Wireshark**

### 3.4.9 Install GitHub Copilot extension (or other AI assistant)

AI assistants are not necessary for DevCon sessions, but they will be used to perform some customizations. Presenters will highlight potential advantages and tips-and-tricks using AI tools.

#### 3.4.9.1 Install GitHub Copilot extension for Infineon employees

1. Instructions to install GitHub Copilot are available at <http://goto/copilot>
  - a. Read instructions at Getting Started:  
<https://confluencewikiprod.intra.infineon.com/spaces/DEVSTACK/pages/1798207560/2.+Copilot+-+Getting+Started>
  - b. Request GitHub Copilot license:  
<https://confluencewikiprod.intra.infineon.com/spaces/DEVSTACK/pages/2219312433/2.1+Copilot+-+Get+Access+to+Github+Copilot>
  - c. Set up Copilot for VSCode:  
<https://confluencewikiprod.intra.infineon.com/spaces/DEVSTACK/pages/2221394955/2.2.1+Copilot+-+VS+Code>
2. Confirm installation by opening a Copilot chat in VS Code (Ctrl + Shift + i)



**Figure 19 Copilot chat in VSCode**

### 3.4.9.2 Install AI assistant for non-Infineon employees

1. Follow your organization's guidelines for installing AI assistant tools or request a personal license.

## 3.5 Additional tools for Zephyr

### 3.5.1 Install Zephyr dependencies

1. Download prework bundle from <https://github.com/luisreynosolFX/mtb-training-devcon-americas-2026/tree/main/zephyr>

The exact destination folder doesn't matter — anywhere you can find again (e.g. Downloads or Desktop) is fine, as long as setup-zephyr-windows-deps.ps1 and gperf-dtc-windows-bundle.zip end up in the same folder after extracting.

Note: the ZIP file includes a README with this instructions and additional troubleshooting details.

2. Open a regular (non-admin) PowerShell window and navigate into that folder (wherever you extracted or cloned it), e.g.

➤ `cd zephyr-windows-training-prereqs`

3. Run the following command:

➤ `powershell -ExecutionPolicy Bypass -File .\setup-zephyr-windows-deps.ps1`

The script downloads each tool's official vendor package, installs everything to your user profile, wires up your **User** PATH, and prints a pass/fail check for all 8 tools. Safe to re-run — it skips steps that are already done.





4. Confirm the scripts worked. The last thing the script prints is a verification table, one line per tool:

**Code listing 1     Output from Zephyr scripts**

```
==> Verifying installed tools
[OK] cmake      cmake version 4.4.2
[OK] ninja      1.13.2
[OK] python      Python 3.12.10
[OK] git         git version 2.55.0.windows.5
[OK] wget       GNU Wget 1.21.4 built on mingw32.
[OK] 7z
[OK] gperf       GNU gperf 3.3
[OK] dtc         Version: DTC 1.7.2
```

**All 8 lines showing [OK] means you're done — no further action needed.**

If any line instead shows [MISSING], close and reopen your terminal (PATH changes only take effect in a new window) and re-run the script once — it will skip everything already installed and only retry what's missing. If a tool still shows [MISSING] after that, see the Troubleshooting section in the README included in the ZIP file.

## 4 Pre-requisite courses to attend the DEVCON training

### 4.1 PSOC™ Control

#### 4.1.1 E3 Power Suite

- E2 Training attendance on PSC3M/P5 given previous years (2024 or 2025)
- alternatively do self-learning leveraging the training available on Infineon Academy at the following link:

<https://catalog.ap-sg-1.icp.infineon.com/pal-catalog?language=1&query=PSOC%E2%84%A2+Control+C3+Session>

#### 4.1.2 PSC3M6 Delta

- E2 Training attendance on PSC3M/P5 given previous years (2024 or 2025)
- alternatively do self-learning leveraging the training available on Infineon Academy at the following link:

<https://catalog.ap-sg-1.icp.infineon.com/pal-catalog?language=1&query=PSOC%E2%84%A2+Control+C3+Session>

#### 4.1.3 PSOC Control C3 Motor Training

- E2 Training attendance on PSC3M/P5 given previous years (2024 or 2025)
- alternatively do self-learning leveraging the training available on Infineon Academy at the following link:

<https://catalog.ap-sg-1.icp.infineon.com/pal-catalog?language=1&query=PSOC%E2%84%A2+Control+C3+Session>

- Also, it is not mandatory but for colleagues who attended motor training last time can do a refresher : self-learning leveraging the training available on Infineon Academy at the following link:

<https://infineon-academy.csod.com/ui/lms-learning-details/app/material/43b370a4-bcfb-4d49-acb9-b24ba7b3f140>

### 4.2 PSOC™ Edge

#### 4.2.1 Introduction to PSOC™ Edge

- Complete self-learning session.
  - Available at [github](#), with video at [Infineon Academy](#).

#### 4.2.2 Getting started with the development ecosystem for PSOC™ Edge E84

- Complete Lab #1 (hello world) using VSCode.
  - Self-learning session (using Eclipse) available at [github](#), with video at [Infineon Academy](#).
  - Updated lab manual using VSCode available at <https://luisreynosoifx.github.io/mtb-training-psoc-edge-e84-intro-ecosystem/pse84-introductory-ecosystem-e2-training-manual.html>